

A-level

# ENVIRONMENTAL SCIENCE

## Paper 2 - Generated practice paper Set 13

### Mark scheme

120 marks: Questions 1-10 (95 marks) plus one essay from Question 11 (25 marks).

### Marking guidance

- Marking points are separated by bullets; each bullet is worth 1 mark unless stated.
- Where a question asks for a fixed number of points (eg *two* reasons), credit only the first points given.
- Alternative correct wording and other relevant, accurate points are credited.
- For levels-of-response questions, first determine the level using the descriptors, then position the mark within the level using the indicative content.
- Calculations: credit correct working (method marks) even if the final answer is wrong; incorrect answers with no working score 0.

**Textbook references:** every question below cites the printed page numbers in R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018) from which the content is drawn, together with the AQA 7447 specification reference.

| Question | Marking guidance                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | Marks |
|----------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|
| 01.1     | <ul style="list-style-type: none"> <li>• Direct measurements of past conditions cannot be taken, so indirect evidence that is related to the condition must be used</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                    | 1     |
| 01.2     | <ul style="list-style-type: none"> <li>• Age / date of the sample (radioisotope dating)</li> <li>• Past temperatures</li> <li>• Composition of the past atmosphere (eg CO<sub>2</sub> / methane concentrations)</li> <li>• Temperature: more magnesium is incorporated into calcite at higher temperatures</li> </ul>                                                                                                                                                                                                                                                                             | 4     |
| 01.3     | <ul style="list-style-type: none"> <li>• Movement of continents changed the position of land masses / latitude and therefore the climate and habitats present</li> <li>• Ocean currents redistribute heat and nutrients; changes in currents (eg as continents moved) altered climates, so past conditions cannot be explained without knowing how the systems worked</li> </ul>                                                                                                                                                                                                                  | 2     |
| 01.4     | <p><i>Any 3 from:</i></p> <ul style="list-style-type: none"> <li>• There was no oxygen / ozone layer so high levels of UV radiation reached the land surface; water absorbed the UV and protected organisms in the oceans</li> <li>• Warm temperatures around the vents and a rich mix of chemicals / increasingly complex organic molecules in the sea made biological reactions more likely</li> <li>• The atmosphere contained toxic gases such as ammonia; the early Archaea were anaerobic so did not need oxygen</li> <li>• Water acted as the solvent for reactions / transport</li> </ul> | 3     |
|          | <p><b>Source:</b> R. Genn, Environmental Science A level AQA (Insight &amp; Perspective, 2018), pp. 6-9; AQA spec 3.1.1.1-3.1.1.3.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                            |       |
| 02.1     | <ul style="list-style-type: none"> <li>• Cross-bred offspring are healthier / more productive because the problems caused by homozygous recessive genes in inbred varieties are reduced (heterosis)</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                    | 1     |

| Question | Marking guidance                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | Marks |
|----------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|
| 02.2     | <ul style="list-style-type: none"> <li>• Zebu tolerate heat but have a low milk yield; Ayrshire have a high milk yield but are not heat tolerant</li> <li>• Cross-breeding combined the desirable characteristics: heat tolerance with a high milk yield</li> </ul>                                                                                                                                                                                                                                                                                                                             | 2     |
| 02.3     | <p><i>Any 2 from:</i></p> <ul style="list-style-type: none"> <li>• Replacement of valuable animals that die with genetically identical individuals</li> <li>• Replacement of herds culled during a disease outbreak</li> <li>• Production of large numbers of individuals with desirable characteristics</li> </ul>                                                                                                                                                                                                                                                                             | 2     |
|          | <b>Source:</b> R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 321-322; AQA spec 3.5.1.2.                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |       |
| 03.1     | <ul style="list-style-type: none"> <li>• Rotation: each crop has its own pests; growing a different crop each year (4-5 year cycle) means pests left at the end of a season die before that crop returns, so populations cannot build up early</li> <li>• Beetle banks / hedgerows provide habitat and overwintering sites for natural predators (ground beetles, ladybirds) that eat aphids</li> <li>• Rotation also evens out nutrient demands and includes legumes</li> </ul>                                                                                                                | 4     |
| 03.2     | <ul style="list-style-type: none"> <li>• Integrated control gives almost the same yield as chemical control (18.1 vs 18.5) with far fewer applications (1-2 vs 6)</li> <li>• It retains most of the biodiversity of cultural control (26 vs 31 species) whereas chemical control alone leaves only 8</li> <li>• Cultural control alone loses about 12% yield; integrated control combines cultural methods, resistant varieties and carefully timed specific non-persistent pesticides only when essential - best balance of yield, cost and impact</li> </ul>                                  | 3     |
| 03.3     | <ul style="list-style-type: none"> <li>• Large numbers of males are sterilised with gamma radiation and released</li> <li>• Females of many insects mate only once and store sperm, so mating with a sterile male produces no offspring; enough sterile males reduce fertile matings below replacement level (screw worm fly, Mediterranean fruit fly)</li> <li>• Only the target species' females mate with them so non-target species are unaffected; fails if sterilised males behave abnormally or re-colonisation occurs</li> </ul>                                                        | 3     |
|          | <b>Source:</b> R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 314-318; AQA spec 3.5.1.1.                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |       |
| 04.1     | <ul style="list-style-type: none"> <li>• Demersal trawl: seabed species - cod, haddock, plaice, shrimps, scampi; chains / weights damage the seabed (mix aerobic and anaerobic layers, destroy sea fans, corals, seagrass); high by-catch in mixed fisheries; high fuel use</li> <li>• Purse seine: shoaling pelagic species - tuna, sardines, herring, anchovy; can remove whole shoals; dolphins associated with tuna shoals may be caught</li> <li>• Pelagic long-line: tuna, squid; albatrosses (long-lived, low breeding rate) drown on baited hooks; sharks and turtles caught</li> </ul> | 6     |
| 04.2     | <ul style="list-style-type: none"> <li>• Shrimp trawls use very small mesh so crabs, molluscs, sea urchins, starfish, juvenile fish cannot escape - by-catch can be 99% of the catch</li> <li>• Drift nets are non-selective curtains near the surface catching whales, dolphins, turtles, sharks</li> </ul>                                                                                                                                                                                                                                                                                    | 2     |

| Question    | Marking guidance                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | Marks |
|-------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|
| <b>04.3</b> | <ul style="list-style-type: none"> <li>• Lost or discarded nets and traps continue to catch and kill organisms; dead animals act as bait attracting more</li> <li>• Biodegradable ropes / panels so lost traps fall apart; radio transmitters on traps so they can be recovered; net recovery schemes</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | 2     |
|             | <b>Source:</b> R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 340-343; AQA spec 3.5.2.2.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |       |
| <b>05.1</b> | <ul style="list-style-type: none"> <li>• Carnivores fed pellets of sandeels, herring, anchovy; overfishing sandeels reduces puffins and terns; reduced by more plant protein / vegetable oil in pellets or farming herbivores (tilapia)</li> <li>• High density spreads lice (which spread to wild fish) and infections; pesticides (organophosphates, pyrethroids, hydrogen peroxide) kill wildlife; antibiotics breed resistant bacteria; reduced by lower densities, single-age tanks with flow from young to old, wrasse as biological control, warm-water washes, circulating flow</li> <li>• Escaped farmed fish breed with wild fish introducing disadvantageous genes (bright colour), or non-indigenous escapees colonise as predators / competitors; reduced by better cage design, sterile triploid fish (pressure-treated eggs), species that cannot breed locally (Japanese oysters in UK)</li> <li>• Faeces and uneaten food decompose causing deoxygenation under cages; reduced by monitored feeding, siting cages in strong currents, effluent treatment / reed beds for freshwater farms, IMTA with seaweed and shellfish absorbing wastes</li> </ul> | 8     |
| <b>05.2</b> | <ul style="list-style-type: none"> <li>• Predators are killed under licence to protect stock, reducing wild populations of protected / valued species</li> <li>• Better cage designs, netting, fencing, bird scarers exclude predators without killing</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | 2     |
|             | <b>Source:</b> R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 356-359; AQA spec 3.5.2.3.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |       |
| <b>06.1</b> | <ul style="list-style-type: none"> <li>• As sea otters declined (1960-1980) sea urchins increased (to 350) and kelp area fell (to 25)</li> <li>• As otters recovered after 1990 urchins declined and kelp recovered</li> <li>• Otter and kelp trends are positively correlated; urchins negatively correlated with both</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | 3     |
| <b>06.2</b> | <ul style="list-style-type: none"> <li>• Sea otters are predators of sea urchins so fewer otters allowed the urchin population to rise</li> <li>• Sea urchins graze kelp so the kelp forest was over-grazed and destroyed</li> <li>• Kelp forests are habitat / food for many species including otters (keystone species / trophic cascade); otter recovery restored the balance</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | 3     |
| <b>06.3</b> | <p><i>Any 2 from:</i></p> <ul style="list-style-type: none"> <li>• Over-fishing of sandeels reduced puffins</li> <li>• Over-collection of turtle eggs reduced turtles so jellyfish (their food) increased; over-fishing of sardines / herring (competitors) also increased jellyfish</li> <li>• Loss of pollinating bees (pesticides, wild flower loss) reduces plant reproduction; loss of elephants / large herbivores reduces seed dispersal of trees</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | 2     |
| <b>06.4</b> | <ul style="list-style-type: none"> <li>• They control abiotic features: beaver dams create ponds colonised by aquatic species; elephants create clearings and keep water holes open</li> <li>• Species that depend on these features (light, water, minerals) cannot survive in their absence</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | 2     |

| Question    | Marking guidance                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | Marks |
|-------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|
|             | <b>Source:</b> R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 23-25; AQA spec 3.1.2.2.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |       |
| <b>07.1</b> | <ul style="list-style-type: none"> <li>• Small clearings were farmed for a few years then abandoned as nutrients declined; the forest recolonised and soil fertility recovered before the area was cleared again</li> <li>• Population growth shortens the gap between clearances so the forest and soil do not recover</li> <li>• Clearings enlarge and degrade; less regrowth</li> </ul>                                                                                                                                                                                                                                                                | 3     |
| <b>07.2</b> | <ul style="list-style-type: none"> <li>• Rainforest soils are thin and nutrient-poor (nutrients held in the biomass); once cleared, fertility falls in a few years and yields drop, so more land is cleared</li> <li>• Abandoned land may not be recolonised if no forest survives nearby; erosion of exposed soil</li> </ul>                                                                                                                                                                                                                                                                                                                             | 2     |
| <b>07.3</b> | <ul style="list-style-type: none"> <li>• Mining (aluminium, iron, gold, copper) destroys habitat directly and pollutes rivers with turbid drainage, cyanide and mercury from processing</li> <li>• Reservoirs (eg Tucurui, 2850 km<sup>2</sup>) flood large areas, block migratory fish and river dolphins; decomposing vegetation produces acidic, deoxygenated water; fluctuating downstream flows</li> <li>• Droughts increase frequency, area and duration of fires (long fires boil sap and kill trees); cloud forests lose humidity as clouds form higher so frogs and epiphytes decline; some models predict replacement by savannah</li> </ul>    | 3     |
| <b>07.4</b> | <ul style="list-style-type: none"> <li>• A creditor country or conservation charity cancels / buys part of a debtor country's foreign debt</li> <li>• In return the country guarantees to protect an area of rainforest (eg USA - Guatemala, Brazil; WWF - Costa Rica, Philippines)</li> </ul>                                                                                                                                                                                                                                                                                                                                                            | 2     |
|             | <b>Source:</b> R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 55-57; AQA spec 3.1.2.3.4.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |       |
| <b>08.1</b> | <ul style="list-style-type: none"> <li>• Other MEDCs (USA, Canada, Australia) also had surpluses; Japan already had suppliers</li> <li>• Eastern Europe / USSR needed food but could not pay</li> <li>• Selling cheaply in LEDCs would undercut local farmers, put them out of business and reduce their long-term food production</li> </ul>                                                                                                                                                                                                                                                                                                             | 3     |
| <b>08.2</b> | <ul style="list-style-type: none"> <li>• Quotas limiting production (eg milk)</li> <li>• Set-aside payments for taking land out of production</li> <li>• Farm diversification (recreation, farm shops, specialised products) and alternative crops / livestock (biofuels, pharmaceutical poppies, llamas, deer, bees)</li> <li>• Agri-environmental schemes (ESA, CSS, ESS, CS) paying for environmentally beneficial farming</li> </ul>                                                                                                                                                                                                                  | 4     |
| <b>08.3</b> | <p><i>Any 3 from:</i></p> <ul style="list-style-type: none"> <li>• Hedgerows at least 1.5 m tall cut at 2-year intervals outside the breeding season - bird / insect habitat and landscape</li> <li>• Buffer strips / field margins (no cultivation, fertiliser or pesticide within 2 m) - protect ditches and rivers, habitat</li> <li>• Beetle banks of tussocky grass - natural pest predators</li> <li>• Skylark nesting plots; wild bird seed / nectar plants - winter food and pollinators</li> <li>• In-field tree protection; controlled stocking density to reduce compaction; ditch management; dry-stone walls; woodland management</li> </ul> | 3     |

| Question    | Marking guidance                                                                                                                                                                                                                                                                                                                                                                                                                                     | Marks |
|-------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|
|             | <b>Source:</b> R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 332-334; AQA spec 3.5.1.4.                                                                                                                                                                                                                                                                                                                              |       |
| <b>09.1</b> | <ul style="list-style-type: none"> <li>• Anchovy r-selected; blue whale k-selected</li> <li>• r-selected: mature quickly, produce many young, disperse widely, short lives; adapted to respond rapidly to high death rates / unstable conditions (mice, locusts, greenfly, rabbits)</li> <li>• k-selected: mature late, few young with parental care, long-lived; populations stable near carrying capacity (whales, elephants, rhinos)</li> </ul>   | 3     |
| <b>09.2</b> | <ul style="list-style-type: none"> <li>• Low reproductive rate means losses from hunting / habitat change cannot be replaced quickly</li> <li>• An increase in death rate above the low birth rate causes a population crash</li> <li>• Recovery takes decades (eg blue whale still far below pre-whaling numbers after the 1986 moratorium); r-selected species rebound within a few years once exploitation stops</li> </ul>                       | 3     |
| <b>09.3</b> | <ul style="list-style-type: none"> <li>• New population = starting population + births + immigrants - deaths - emigrants</li> </ul>                                                                                                                                                                                                                                                                                                                  | 1     |
| <b>09.4</b> | <ul style="list-style-type: none"> <li>• The greatest rate of exploitation (harvest) possible without causing long-term population decline</li> <li>• Equals the surplus production: highest when the population is well below carrying capacity where growth rate is fastest (middle of the sigmoid curve)</li> <li>• At carrying capacity net growth is zero so any harvest reduces the population; harvesting above MSY causes decline</li> </ul> | 3     |
|             | <b>Source:</b> R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 79-80; AQA spec 3.1.3.4.                                                                                                                                                                                                                                                                                                                                |       |

| Question | Marking guidance                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | Marks |
|----------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|
| 10.1     | <p><b>Levels of response</b></p> <p><b>Level 3 (7-9 marks):</b> Answer is comprehensive, relevant and logically structured. Shows detailed knowledge and understanding of the relevant scientific ideas and applies them accurately to the context. Evaluation/analysis is well supported by evidence and a clear judgement or conclusion is reached.</p> <p><b>Level 2 (4-6 marks):</b> Answer is mostly relevant and shows some structure. Knowledge and understanding of relevant ideas is sound but application to the context or evaluation may be incomplete or lack detail. A judgement may be made but is not fully supported.</p> <p><b>Level 1 (1-3 marks):</b> Answer is basic, with limited structure. Some relevant knowledge is shown but there are omissions or errors. Little application to the context and little or no evaluation.</p> <p><b>0 marks:</b> No relevant content.</p> <p><b>Indicative content</b> (credit other relevant, accurate points)</p> <ul style="list-style-type: none"> <li>• In-situ: species remain in natural conditions with the whole community and inter-species relationships; protected areas (SSSI, NNR, SAC, SPA, Ramsar, MCZ), land ownership, agri-environment schemes, habitat creation and design (area, shape, corridors, diversity), management of abiotic and biotic factors, plagioclimax maintenance, population control, reintroductions and rewilding (Yellowstone wolves, Scottish beavers)</li> <li>• Limits of in-situ: external threats (pollution, climate change), enforcement, cost, conflicts with land use, small fragmented populations</li> <li>• Ex-situ: captive breeding when in-situ is impossible or insufficient; selection criteria; difficulties keeping and breeding (habitat size, food, relationships, breeding triggers, group dynamics, gene pool / inbreeding, hybridisation); techniques (stud books, cryopreservation, AI, embryo transfer, micropropagation, cloning); seed banks; successes (Hawaiian goose, Iberian lynx, red kite, large blue)</li> <li>• Limits of ex-situ: expensive, few species, loss of wild behaviour, small gene pools, release requires suitable habitat and post-release support (hard vs soft release, monitoring)</li> <li>• Judgement: ex-situ is a temporary safeguard / support that only succeeds if in-situ habitat exists to return to; both needed, in-situ fundamental</li> </ul> | 9     |
| 10.2     | <ul style="list-style-type: none"> <li>• Restoring natural processes / communities to a habitat changed by humans, often by reintroducing missing (keystone) species and removing human management such as grazing</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | 1     |
|          | <p><b>Source:</b> R. Genn, Environmental Science A level AQA (Insight &amp; Perspective, 2018), pp. 32-49; AQA spec 3.1.2.3.3-3.1.2.3.4.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |       |

| Question | Marking guidance                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | Marks |
|----------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|
| 11.1     | <p><b>Discuss how ecological monitoring and the development of new technologies have improved the planning and evaluation of conservation strategies.</b></p> <p><b>Levels of response</b></p> <p><b>Level 5 (21-25 marks):</b> Comprehensive and detailed knowledge and understanding drawn from across the specification, applied accurately to the question. Argument is coherent, well-structured and supported by a wide range of relevant, well-chosen examples and quantitative detail where appropriate. Interconnections between topics are made explicitly and a clear, justified conclusion is reached.</p> <p><b>Level 4 (16-20 marks):</b> Detailed knowledge and understanding with good application to the question. Argument is logically organised with relevant examples; some links between different areas of the specification are made. Conclusions are drawn but may be less fully justified.</p> <p><b>Level 3 (11-15 marks):</b> Sound knowledge and understanding, mostly applied to the question. Some structure and relevant examples, but coverage may be uneven or descriptive rather than analytical. Limited linking of topics.</p> <p><b>Level 2 (6-10 marks):</b> Basic knowledge and understanding with partial relevance. Answer is largely descriptive with few examples and limited structure; some errors or omissions.</p> <p><b>Level 1 (1-5 marks):</b> Limited, fragmented knowledge with little relevance to the question. Little structure; examples absent or inappropriate.</p> <p><b>0 marks:</b> No relevant content.</p> <p><b>Indicative content</b> (students are not expected to cover all of these; credit other relevant material)</p> <ul style="list-style-type: none"> <li>• Why monitor: species present, population size, density, distribution, survival, age structure, breeding rates, movements; baseline and evaluation; Red List categorisation and quotas rely on it</li> <li>• Standard techniques and their limitations: quadrats, transects, nets, traps, Lincoln index, Simpson's index, DAFOR; sampling design (random / systematic, size, number, timing, standardisation)</li> <li>• New technologies: satellite / radio / GPS / acoustic tracking, geolocators, camera traps, CCTV, image recognition databases, DNA databases, eDNA, sonograms, drones / ROVs / AUVs, satellite sensors, data loggers</li> <li>• How they improve validity: representative data over large areas and times, less disturbance, individual identification, detection of hidden species, objectivity</li> <li>• Examples: penguin colonies from satellites, tiger camera traps, ivory DNA, MCZ monitoring, released animals tracked</li> <li>• Limitations: cost, battery / tag weight, data handling, ethics, still need ground truthing</li> <li>• Judgement</li> </ul> <p><b>Source:</b> R. Genn, Environmental Science A level AQA (Insight &amp; Perspective, 2018), pp. 393-394, 400-407; AQA spec 3.1.2.3.5-3.1.2.3.6, 3.7.</p> | 25    |

| Question | Marking guidance                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | Marks |
|----------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|
| 11.2     | <p><b>Discuss the extent to which the principles of the circular economy could be used to develop sustainable lifestyles.</b></p> <p><b>Levels of response</b></p> <p><b>Level 5 (21-25 marks):</b> Comprehensive and detailed knowledge and understanding drawn from across the specification, applied accurately to the question. Argument is coherent, well-structured and supported by a wide range of relevant, well-chosen examples and quantitative detail where appropriate. Interconnections between topics are made explicitly and a clear, justified conclusion is reached.</p> <p><b>Level 4 (16-20 marks):</b> Detailed knowledge and understanding with good application to the question. Argument is logically organised with relevant examples; some links between different areas of the specification are made. Conclusions are drawn but may be less fully justified.</p> <p><b>Level 3 (11-15 marks):</b> Sound knowledge and understanding, mostly applied to the question. Some structure and relevant examples, but coverage may be uneven or descriptive rather than analytical. Limited linking of topics.</p> <p><b>Level 2 (6-10 marks):</b> Basic knowledge and understanding with partial relevance. Answer is largely descriptive with few examples and limited structure; some errors or omissions.</p> <p><b>Level 1 (1-5 marks):</b> Limited, fragmented knowledge with little relevance to the question. Little structure; examples absent or inappropriate.</p> <p><b>0 marks:</b> No relevant content.</p> <p><b>Indicative content</b> (students are not expected to cover all of these; credit other relevant material)</p> <ul style="list-style-type: none"> <li>• Definition of sustainability (Brundtland); population x per capita impact; dynamic equilibria, feedback, tipping points, diversity</li> <li>• Principles of natural systems: renewable low-density energy, low temperatures, material cycles, non-toxic wastes; contrast with linear human systems</li> <li>• Circular economy principles: cycling materials, renewable energy, supporting ecosystems, separation of technical and biological materials, diversity, connected systems, design for extended use and re-use, optimum production, new technologies</li> <li>• Application to land use, water, minerals, waste, pollution, energy, food with examples (Kalundborg, Eigg, Ol Pejeta, Rio Bravo, EU initiatives, DC grids)</li> <li>• Measuring progress: ecological footprint, biocapacity, global hectares, Earth Overshoot Day, carbon footprint, Living Planet Index</li> <li>• Barriers: economics, competition, consumer behaviour, need for international agreement, existing infrastructure</li> <li>• Judgement on the extent achievable</li> </ul> <p><b>Source:</b> R. Genn, Environmental Science A level AQA (Insight &amp; Perspective, 2018), pp. 370-389; AQA spec 3.6.</p> | 25    |